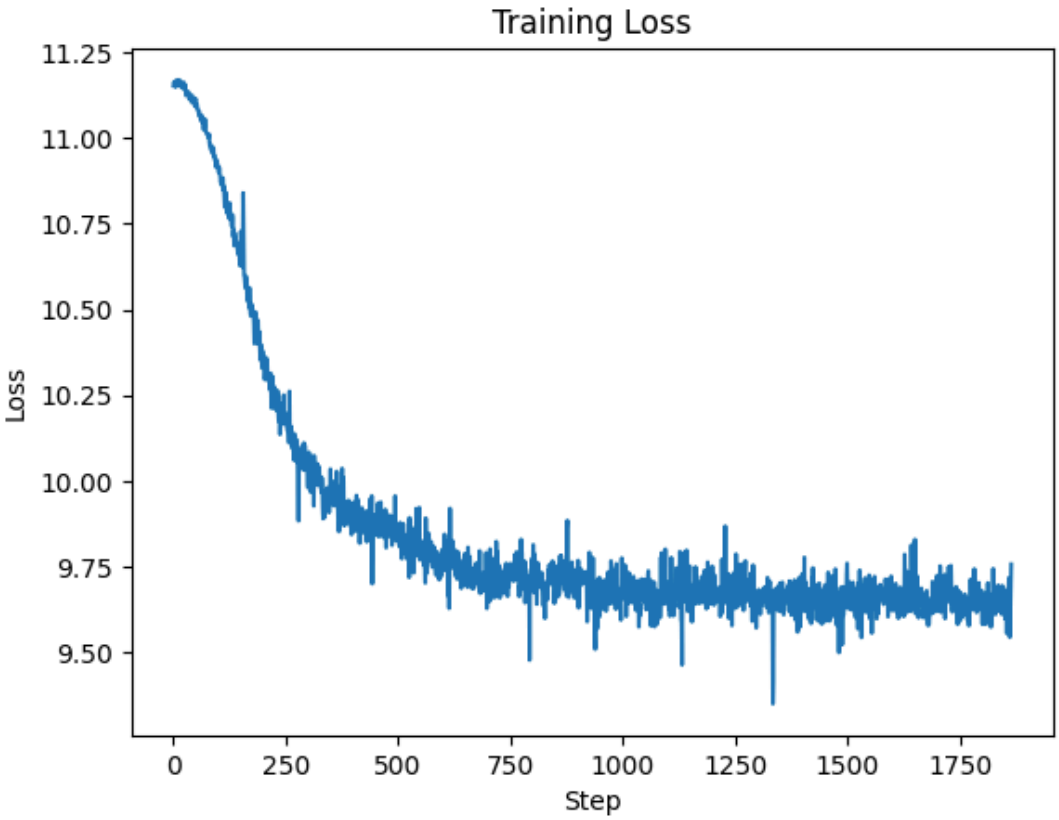
BLUBRIDGE

Anuj Kumar <anuj.kumar@blubridge.com>

[EOD REPORT] Anuj Kumar - 17/02/2026

2 messages

Anuj Kumar <anuj.kumar@blubridge.com> 17 February 2026 at 20:00
To: Manoj Kumar <manoj@blubridge.com>
Cc: Praveen Kumar <praveen@blubridge.com>, Kripa Shankar <kripa@blubridge.com>, Adhitya Charan <adhitya.charan@blubridge.com>, Kiruthik Kanna M <kiruthik.kanna@blubridge.com>, Navin Kumar V <navin.kumar@blubridge.com>, Dinesh G <dinesh.g@blubridge.com>



Tweaked the gradient clipping function and got this curve.

Although I got only ~30k throughput.

Decided to pull the master branch.

Initially got a very low throughput, with a 24M model only 14000 tok/sec.

With a 96m model, only 7000.

```
Total parameters: 96062976
Parameter Split:
  Decay Params (Weights): 24
  No-Decay Params (Bias/LN): 28
DataLoader: found 3 shards for split train
DataLoader: found 1 shards for split val
step     0 | loss: 11.276883 | lr 3.2258e-06 | norm: 2.6468 | dt: 9428.81ms | tok/sec: 6950.61 | Tensors: 128 | Mem: 2218MB
step     1 | loss: 11.255655 | lr 6.4516e-06 | norm: 2.7301 | dt: 9249.34ms | tok/sec: 7085.48 | Tensors: 128 | Mem: 2218MB
step     2 | loss: 11.250415 | lr 9.6774e-06 | norm: 2.8444 | dt: 9266.22ms | tok/sec: 7072.57 | Tensors: 128 | Mem: 2218MB
step     3 | loss: 11.151780 | lr 1.2903e-05 | norm: 2.7203 | dt: 9262.36ms | tok/sec: 7075.52 | Tensors: 128 | Mem: 2218MB
step     4 | loss: 11.101754 | lr 1.6129e-05 | norm: 2.6734 | dt: 9290.76ms | tok/sec: 7053.89 | Tensors: 128 | Mem: 2218MB
step     5 | loss: 10.992039 | lr 1.9355e-05 | norm: 2.7427 | dt: 9387.33ms | tok/sec: 6981.33 | Tensors: 128 | Mem: 2218MB
step     6 | loss: 10.891563 | lr 2.2581e-05 | norm: 2.6750 | dt: 9341.82ms | tok/sec: 7015.34 | Tensors: 128 | Mem: 2218MB
step     7 | loss: 10.774196 | lr 2.5806e-05 | norm: 2.6168 | dt: 9376.09ms | tok/sec: 6989.70 | Tensors: 128 | Mem: 2218MB
^Cblu-bridge005@blubridge005-MS-7E06:~/Desktop/Anuj@BluBridge/Parallelism/Tensor Parallelism/beta/DTensor_v2.0/mlp$ cd '/home
```

Decided to break down the timings for each step.

```
Total parameters: 96062976
Parameter Split:
  Decay Params (Weights): 24
  No-Decay Params (Bias/LN): 28
DataLoader: found 3 shards for split train
DataLoader: found 1 shards for split val
step     0 | loss: 11.2769 | tok/s: 6744 | ms: 9717.0 [Data:0 Fwd:5726 Bwd:3569 Sync:0 Clip:307 Opt:16] | Mem: 2218MB
step     1 | loss: 11.2558 | tok/s: 6926 | ms: 9461.9 [Data:0 Fwd:5386 Bwd:3594 Sync:0 Clip:309 Opt:8] | Mem: 2218MB
step     2 | loss: 11.2507 | tok/s: 6989 | ms: 9377.0 [Data:0 Fwd:5342 Bwd:3542 Sync:0 Clip:317 Opt:9] | Mem: 2218MB
step     3 | loss: 11.1520 | tok/s: 6978 | ms: 9392.0 [Data:0 Fwd:5313 Bwd:3602 Sync:0 Clip:311 Opt:8] | Mem: 2218MB
step     4 | loss: 11.1018 | tok/s: 6881 | ms: 9524.0 [Data:0 Fwd:5386 Bwd:3631 Sync:0 Clip:323 Opt:8] | Mem: 2218MB
step     5 | loss: 10.9919 | tok/s: 6952 | ms: 9427.0 [Data:0 Fwd:5324 Bwd:3616 Sync:0 Clip:313 Opt:9] | Mem: 2218MB
step     6 | loss: 10.8916 | tok/s: 6951 | ms: 9428.9 [Data:0 Fwd:5324 Bwd:3615 Sync:0 Clip:311 Opt:9] | Mem: 2218MB
step     7 | loss: 10.7742 | tok/s: 6994 | ms: 9370.8 [Data:0 Fwd:5293 Bwd:3608 Sync:0 Clip:300 Opt:8] | Mem: 2218MB
^Cstep     8 | loss: 10.6766 | tok/s: 7051 | ms: 9293.9 [Data:0 Fwd:5293 Bwd:3521 Sync:0 Clip:311 Opt:9] | Mem: 2218MB
```

The forward pass was slower than expected, considering the backward pass which was 3500 ms, I expected a forward pass of 1700 ms.

Did a proper breakdown of the forward pass, and found the culprit to be cross_entropy_loss function.

CrossEntropyLoss alone was taking about 140ms every time it was called.

Subtraction Latency: Dropped from 118ms to ~8ms (14x faster).

Total CE Loss Latency: Dropped from 140ms to ~25ms (5.6x faster).

Finally reduced the time.

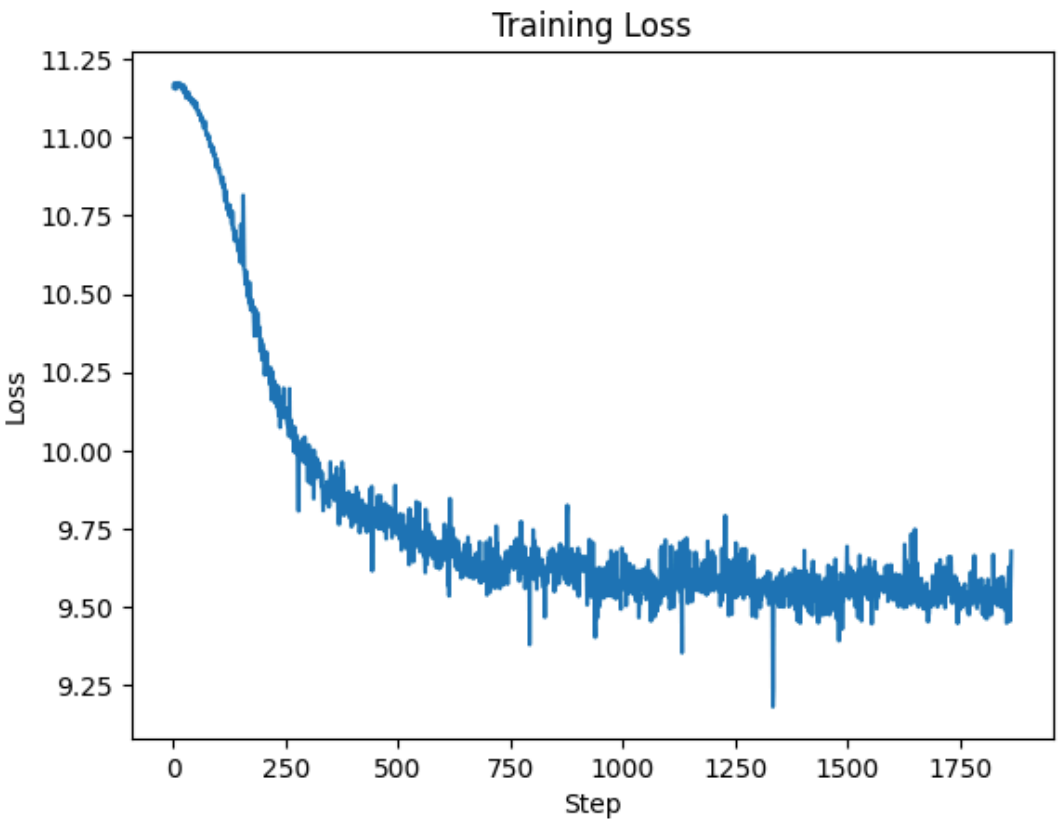
```
Total parameters: 24432384
Parameter Split:
  Decay Params (Weights): 8
  No-Decay Params (Bias/LN): 12
DataLoader: found 3 shards for split train
DataLoader: found 1 shards for split val
step     0 | loss: 11.1606 | tok/s: 56488 | ms: 1160.2 [Data:0 Fwd:727 Bwd:400 Sync:0 Clip:30 Opt:1] | Mem: 216MB
step     1 | loss: 11.1691 | tok/s: 69139 | ms: 947.9 [Data:0 Fwd:505 Bwd:396 Sync:0 Clip:31 Opt:1] | Mem: 216MB
step     2 | loss: 11.1576 | tok/s: 68620 | ms: 955.1 [Data:0 Fwd:504 Bwd:403 Sync:0 Clip:31 Opt:1] | Mem: 216MB
step     3 | loss: 11.1478 | tok/s: 68869 | ms: 951.6 [Data:0 Fwd:502 Bwd:401 Sync:0 Clip:32 Opt:1] | Mem: 216MB
step     4 | loss: 11.1390 | tok/s: 69777 | ms: 939.2 [Data:0 Fwd:498 Bwd:395 Sync:0 Clip:30 Opt:1] | Mem: 216MB
step     5 | loss: 11.1034 | tok/s: 69391 | ms: 944.4 [Data:0 Fwd:498 Bwd:399 Sync:0 Clip:32 Opt:1] | Mem: 216MB
step     6 | loss: 11.1032 | tok/s: 69215 | ms: 946.9 [Data:0 Fwd:497 Bwd:402 Sync:0 Clip:31 Opt:1] | Mem: 216MB
step     7 | loss: 11.0821 | tok/s: 69304 | ms: 945.6 [Data:0 Fwd:504 Bwd:395 Sync:0 Clip:30 Opt:1] | Mem: 216MB
step     8 | loss: 11.0500 | tok/s: 69156 | ms: 947.7 [Data:0 Fwd:498 Bwd:402 Sync:0 Clip:31 Opt:1] | Mem: 216MB
step     9 | loss: 11.0176 | tok/s: 69075 | ms: 948.8 [Data:0 Fwd:499 Bwd:401 Sync:0 Clip:31 Opt:1] | Mem: 216MB
step    10 | loss: 10.9967 | tok/s: 69635 | ms: 941.1 [Data:0 Fwd:494 Bwd:402 Sync:0 Clip:29 Opt:1] | Mem: 216MB
step    11 | loss: 10.9761 | tok/s: 69544 | ms: 942.4 [Data:0 Fwd:498 Bwd:399 Sync:0 Clip:29 Opt:1] | Mem: 216MB
step    12 | loss: 10.9245 | tok/s: 70400 | ms: 930.9 [Data:0 Fwd:491 Bwd:395 Sync:0 Clip:29 Opt:1] | Mem: 216MB
step    13 | loss: 10.8657 | tok/s: 70211 | ms: 933.4 [Data:0 Fwd:493 Bwd:394 Sync:0 Clip:30 Opt:1] | Mem: 216MB
step    14 | loss: 10.8256 | tok/s: 70424 | ms: 930.6 [Data:0 Fwd:494 Bwd:390 Sync:0 Clip:30 Opt:1] | Mem: 216MB
step    15 | loss: 10.8067 | tok/s: 70473 | ms: 929.9 [Data:0 Fwd:493 Bwd:391 Sync:0 Clip:29 Opt:1] | Mem: 216MB
step    16 | loss: 10.7592 | tok/s: 70256 | ms: 932.8 [Data:0 Fwd:492 Bwd:394 Sync:0 Clip:30 Opt:1] | Mem: 216MB
step    17 | loss: 10.6814 | tok/s: 70256 | ms: 932.8 [Data:0 Fwd:496 Bwd:391 Sync:0 Clip:29 Opt:1] | Mem: 216MB
```

Right now the total time for a 24M model is ~944 ms per step.

But the forward time is still higher than the backward time.

Yet to pinpoint the culprit for the stall.

Getting this graph as of now -



Regards,
Anuj Kumar

Manoj Kumar Singh <manoj@blubridge.com> 17 February 2026 at 22:55
To: Anuj Kumar <anuj.kumar@blubridge.com>
Cc: Praveen Kumar <praveen@blubridge.com>, Kripa Shankar <kripa@blubridge.com>, Adhitya Charan <adhitya.charan@blubridge.com>, Kiruthik Kanna M <kiruthik.kanna@blubridge.com>, Navin Kumar V <navin.kumar@blubridge.com>, Dinesh G <dinesh.g@blubridge.com>

Nice ok.
[Quoted text hidden]
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Regards,

Manoj Kr. Singh,
CTO
blubridge.png

Blubridge Technologies Pvt.Ltd.
Regs: Plot No. E160 Tiger Varadhachari Road, Kalakshetra Colony,
Besant Nagar, Chennai - 600090.
Email : manoj@blubridge.com **Web** : Blubridge.ai | **Mobile** : 9940429368 | **Phone** : 044-466-00222

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